Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/11/2024
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	Mouse Models of Cancer Risk and Prevention from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	01/29/2016	**End Date:**	12/31/2024
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Key Personnel Changes/Previous PI:	Silvia Siteni, PhD postdoctoral trainee, moved to Milan, Italy 2024; Krishna Luitel (completed PhD) now a Physician Assistant.		
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			

Task Description:

Overall hypothesis: Low-dose radiation induces molecular manifestations of a pro-inflammatory response as a function of radiation type, radiation doses, doses rates, LET (linear energy transfer) value, and time. An oral available anti-inflammatory countermeasure, already in human clinical trials with a good safety profile, will significantly reduce proton and HZE-ion (high charge energy-ion) exposure associated tumor initiation and progression.
The overarching hypothesis for this project is that space radiation induces molecular manifestations of a pro-inflammatory response as a function of radiation type, radiation doses, doses rates, LET value, and time. We are testing if an oral available anti-oxidant and anti-inflammatory countermeasure, already in human clinical trials with a good safety profile, CDDO, significantly reduces proton and HZE-ion exposure associated tumor initiation and progression. Based on experiments conducted at the NASA Space Radiation Laboratory (Brookhaven, NY) we demonstrate that HZE ion components of GCR (galactic cosmic radiation) result in persistent DNA damage and inflammatory signaling, increased mutations in tumor suppressor genes, and higher rates of cancer initiation and progression compared to that seen with similar doses of terrestrial radiation. While physical shielding may reduce some of the risks of space radiation, there is substantial evidence that biological countermeasures will be required to ensure that the established safety limits of increased lifetime fatal cancer risks are not exceeded. We are conducting GCR simulations consisting of fast switching between protons, helium, and silicon using a dose rate of 0.5 cGy/min and a total combined dose of between 27-30 cGy to more closely mimic the space environment on a trip to Mars and back. Finally, we are conducting experiments with the official NASA GCRsim with acute and protracted mixed fields.

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Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Although radiation therapy is commonly used for treatment of many human diseases, including cancer, ionizing radiation produces reactive oxygen species that can damage both cancer and healthy cells in tissues. We have demonstrated using the biological countermeasure, CDDO, an anti-oxidant, anti-inflammatory modulator with a known mechanism of action, a reduction in both cancer initiation and progression in mouse models after exposure to either terrestrial or space radiation. We also demonstrated that CDDO can be used as a radioprotector in normal non cancerous human lung and breast epithelial cells exposed to space and terrestrial irradiations while cancer cells were not protected. This suggests the use of this oral available, non-toxic class of drug can protect non-cancerous healthy cells during radiotherapy, resulting in better outcomes with less toxicity for patients. CDDO has been recently approved by the FDA to treat patients with Friedreich's ataxia. In preliminary experiments aspirin did not provide this protection. Other potential countermeasures such as metformin continue to be tested and show radioprotective activity. In the future, CDDO or metformin may be used to protect astronauts on long-term mission to the Moon or Mars as well as patients receiving radiotherapy on Earth. These radioprotectors may also have utility in protecting first responders to nuclear accidents.

We focused on both acute and protracted full-spectrum galactic cosmic radiation (GCR) simulations. We initiated studies to determine if metformin is an effective radioprotector. Metformin is a biguanide compound used in the treatment of type 2 diabetes mellitus, showing very low cytotoxic effects that was FDA approved in the 1990s and is currently used daily by over 100 million patients with diabetes. Radiotherapy is one of the most efficient non-surgical cancer treatments. Radiotherapy also affects healthy tissues around the tumor, resulting in cellular DNA damage and tissue inflammation increases. Metformin promotes anti-tumor immunity and enhances anti-cancer activity, mainly through the increase of oxidative stress. In contrast, metformin has an opposite effect on healthy tissues, decreasing the oxidative stress and DNA damage in vitro as well as in vivo, by also decreasing chronic inflammation. Metformin acts mainly through the phosphorylation of adenosine monophosphate-activated protein kinase (AMPK), which has pleiotropic effects downstream on cell metabolism. Furthermore, metformin targets mitochondria, inhibiting the complex I of the electron transport chain, but the mechanisms underlying this process have not been completely elucidated. Because of its antioxidant effects, we investigated the role of metformin as a radioprotective compound.
A single dose of metformin (0.5-1mM) on human colon epithelial cells (HCECs) and human skin fibroblasts (BJs), showed an increase in the expression of AMPK alpha subunit phosphorylation and superoxide dismutase 1 (SOD1), during the first 72 hours. Interestingly, SOD1 has a crucial role in protecting against oxidative DNA damage and its overexpression is associated with radio-resistance in human glioma cells. Furthermore, a decrease in the basal DNA damage (phosphorylation of H2AX at Ser 139 foci) and in reactive oxygen species (ROS) production was observed. Moreover, metformin treatment enhanced DNA damage responses (DDR) 24 hours after exposure to 2Gy of γ-rays. To evaluate the radioprotective effect of metformin, cells were treated one time and irradiated 72 hours later, with 2, 4, and 6 Gy doses of gamma-rays. Cells were seeded at low density (200-1000 cells) and a colony formation assay was analyzed after 21 days. Metformin showed an increase in the surviving fraction of cells compared to the irradiated controls. Next, we investigated the radioprotective effect of metformin in vivo. Wild type 129/Sv mice were injected once per day with metformin 200mg/kg, for three consecutive days prior to exposure to 7.5Gy of X-rays and sacrificed

Task Progress:	after 24 hours. Metformin pre-treatment dramatically decreased DNA damage (p53 binding protein 1 foci) in mouse lung and colon tissues as well as the number of micronuclei in bone marrow cells, compared to the irradiated controls. This suggested increased DNA repair. Notably, when mice were irradiated at the dose of 10Gy X-rays post-metformin treatment, an increase of 30% in the surviving fraction was observed. In conclusion, metformin might act as a radioprotector from GCR, potentially lowering the risk of cancer initiation or promotion in astronauts. We have now formulated metformin into lab chow and completed studies at the NASA Space Radiation Laboratory (NSRL) when mice were exposed to 75cGy full spectrum GCR (these are now published in Siteni et al. PLoS One. 2024;19(7): e0307598. https:// Metformin resulted in a hyperpolarization of the mitochondrial membrane, likely induced by the inhibitory effect on the complex I. Furthermore, metformin 0.5mM was able to reduce the reactive oxygen species (ROS) production in combination with 2Gy of γ-rays. Next, the radioprotective effect of metformin in vivo was investigated. Wild type 129/Sv mice were injected once per day with metformin 200mg/kg, for three consecutive days prior to exposure of 7.5Gy of X-rays and sacrificed after 24 hours. Metformin pre-treatment dramatically decreased the DNA damage (p53 binding protein 1 foci) in mouse lung and colon tissues as well as the number of micronuclei in bone marrow cells, compared to the irradiated controls. Notably, when mice were irradiated at the dose of 10 Gy X-rays post-metformin treatment, an increase of 30% in the surviving fraction was observed. Wild type mice were fed with metformin 200mg/kg chow and then irradiated with 2Gy of X-rays and sacrificed after 100 days. Metformin pre-treatment determined the phosphorylation of the protein NF-κB p65 at serine 536, with a major increase in the nuclear fraction. The phosphorylation of the subunit p65 at serine 536 is associated with NF-κB transcriptional activity, but also increased p65 turnover, thus reducing NF-κB activity and supporting the resolution of inflammation. We next tested the GCR spectrum at the NSRL which is composed of 33 beams and uses mixed ion species, protons, and sporadic heavy ions to better simulate the space environment. Wild type mice were fed with 200mg/kg metformin chow for 72 hours and then irradiated with 75cGy GCR. After 4 days, mice were sacrificed and cells from bone marrow were examined. Fixed bone marrow slides were then stained with Giemsa and the number of micronuclei was analyzed. We found that 75cGy GCR increases the number of micronuclei, inducing a genotoxic effect. Administration of metformin 72 hours prior to irradiation decreased the number of micronuclei significantly. Furthermore, DNA damage foci (53BP1) significantly decreased in lung and colon tissues, as well as oxidative stress (8-oxo-G), after 6 hours from irradiation. Metformin pre-administration showed a decrease in apoptosis (cleaved –PARP), and an increase in oxidative DNA damage repair enzyme OGG1 when compared to irradiation only, in colon and lung tissue. Finally, lung cancer susceptible mice K-rasLA-1 were fed with metformin 200mg/kg 4 days prior to 75cGy GCR irradiation and mice were sacrificed after 100 days. Metformin pre-treatment showed a decrease in plasma lipid peroxidation, as well as the number and area of pre-malignant lesions in lung tissue. In conclusion, our published results can be interpreted to suggest that metformin acts as an oral available and safe radioprotector, potentially lowering the risk of cancer initiation or promotion on healthy tissues.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
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